MIL-C-3955C
16 December 1975
SUPERSEDING
MIL-C-3955B
14 August 1967

MILITARY SPECIFICATION

CANS, COMPOSITE, SPIRALLY WOUND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers round, spirally wound composite cans, both single body and telescopic types. The single body type is used in the packaging of parts not exceeding 10 pounds and the telescopic type is used in the packaging of parts not exceeding 20 pounds.

1.2 Classification. Composite cans covered by this specification shall be of the following types, grades, styles, and classes (see 6.2).

Type I - Spirally wound, single body (see Figure 1)

Type II - Spirally wound, telescopic (see Figure 2)

Grade A - Untreated (low moisture resistance, nonwaterproof, not applicable to Type I)

Grade B - Asphalt treated (high moisture resistance and water vapor-proof)

Style a - Inside diameter not exceeding 3 inches

Style b - Inside diameter exceeding 3 inches but not over 7 1/8 inches

Class 1 - Weight capacity up to and including 5 pounds

Class 2 - Weight capacity over 5 pounds up to and including 10 pounds.

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Class 3 - Weight capacity over 10 pounds up to and including 15 pounds (not applicable to Type I)

Class 4 - Weight capacity over 15 pounds up to and including 20 pounds (not applicable to Type I and Type II, grade A)

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

*	QQ-A-1876	-	Aluminum Foil
	QQ-T-191	-	Terne Sheets (Long and Short Terne)
	QQ-T-425	-	Tinplate (Hot Dip and Electrolytic)
	TT-E-485	-	Enamel, Semi-Gloss, Rust Inhibiting
	UU-P-268	-	Paper; Kraft, Wrapping
	MMM-A-260	-	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
	PPP-B-601	-	Boxes, Wood, Cleated-Plywood
	PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
	PPP-B-636	-	Boxes, Shipping, Fiberboard
	PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple Wall

Military

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-A-3029	-	Asphalt, Waterproofing (for Use in Manufacture of Fiber Ammunition Containers)
MIL-A-13374	-	Adhesive, Dextrin (Spiral Tube Winding for Ammunition Containers)
MIL-P-20293	-	Paper, Kraft, Asphalt-Impregnated
MIL-P-20311	-	Paper, Kraft, Duplex, Waterproof Wrapping (for Ammunition Containers)
MIL-B-20390	-	Board, Ammunition Container
MIL-T-43036	-	Tape, Pressure Sensitive Adhesive, Plastic Film, Filament Reinforced (For sealing fiber containers and cans)
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-L-45973	-	Liner Material, Greaseproof

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STANDARDS

Federal		
Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment
Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials, construction and design. The spirally-wound composite cans shall conform to the material, construction and design requirements specified herein.

3.1.1 Materials. Spirally-wound composite cans shall be made from such materials to insure compliance with this specification.

3.1.2 Construction, general. Sides of cans shall be built up, from inside to outside, with plies of specified materials spirally wound in sequence (see figures 3, 4 and 5). All plies shall be bonded together with adhesive made from dextrin conforming to MIL-A-13374 or with animal glues, tackified polyvinyl alcohol adhesive or polyvinyl acetate adhesive meeting performance tests of MIL-A-13374; except where asphalt conforming to MIL-A-3029 or water-resistant adhesive conforming to MMM-A-260 is specified. The ammunition container board shall comply with type I of MIL-B-20390. All plies, except where overlap is specified, shall be wound with butt joints having a maximum gap of 1/16 inch.

3.1.2.1 Metal ends. Metal ends shall be fabricated from terneplate conforming to QQ-T-191, type II, commercial coating weight. Metal ends may also be made of tin plate conforming to QQ-T-425, Class 100, having a coating weight of not less than 1.25 pounds per base box. All metal ends shall be free from cracks or splits. Metal ends shall be primed and painted on the outer side with black or olive-drab enamel conforming to TT-E-485 (color as specified by the procuring agency, see 6.2).

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3.1.2.1.1 Style a cans. Metal ends shall be either flat-center or raised-center type and shall be fabricated from metal conforming to the minimum thickness specified in Table I (see fig. 6).

3.1.2.1.2 Style b cans. Metal ends shall be raised-center type only and shall be fabricated from metal conforming to the minimum thickness specified in Table I (see fig. 6).

TABLE I

Minimum thickness of metal ends

Class	Style a		Style b	
	NOMINAL WT. LB./BASE BOX	THICKNESS INCH	NOMINAL WT. LB./BASE BOX	THICKNESS INCH
1	90	.0097	135	.0149
2	112	.0120	135	.0149
3	112	.0120	135	.0149
4	135	.0149	135	.0149

* 3.1.2.2 Attaching ends. The metal ends shall be applied to the body and closure effected by means of a double crimp (false double seam), as illustrated on figure 7. After crimping, the metal ends shall be smooth, without gathering of metal or paper at the edge of the crimp. The crimped metal ends shall withstand a load applied parallel to the axis of the can in accordance with Table II. Testing shall be as specified in 4.3.3.2.

TABLE II

End stripping strength

Inside diameter of tube	Minimum weight required to strip metal end from tube
Inches	Pounds
Less than 1.50	75
1.50 to 2.37	125
2.38 to 3.37	200
3.38 to 7.125	300

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3.1.3 Construction, specific.

* 3.1.3.1 Type I cans (single-body type). Type I cans shall be constructed as detailed below and in accordance with figures 1 and 3.

3.1.3.1.1 Type I, grade B, style a, class 1 and class 2. Bodies of these cans shall be constructed of six spirally wound plies as specified in 3.1.3.1.1.1 through 3.1.3.1.1.5.

* 3.1.3.1.1.1 First (inner) ply. Material for first ply shall be grease-proof barrier material conforming to MIL-L-45973. The material shall be overlapped not less than 1/4 inch and the overlap glued.

3.1.3.1.1.2 Second and third plies. Material for second and third plies shall be ammunition container board conforming to MIL-B-20390, with a nominal thickness to 0.020 inch.

* 3.1.3.1.1.3 Fourth ply. The fourth ply shall be 0.001 inch minimum thickness, free aluminum foil conforming to QQ-A-1876 overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029, or adhesive conforming to MMM-A-260.

3.1.3.1.1.4 Fifth ply. Material for fifth ply shall be waterproof, duplex kraft paper 0.007 inch thick conforming to MIL-P-20311. The material shall be overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029.

3.1.3.1.1.5 Sixth ply. Material for sixth ply shall be asphalt-impregnated kraft paper 0.006 inch thick conforming to MIL-P-20293. The material shall be overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029. A hairline of asphalt, not more than 1/32 inch in width and 1/64 inch in height, shall be allowable beyond the edge of the outer layer of asphalt-impregnated kraft paper to which the asphalt is applied.

3.1.3.1.2 Type I, grade B, style b, class 1 and class 2. Body of these cans shall be constructed in seven spirally wound plies as specified in 3.1.3.1.2.1 through 3.1.3.1.2.5:

* 3.1.3.1.2.1 First (inner) ply. Material for first ply shall be greaseproof barrier material conforming to MIL-L-45973. The material shall be overlapped not less than 1/4 inch and the overlap glued.

3.1.3.1.2.2 Second, third, and fourth plies. Material for second, third and fourth plies shall be ammunition container board conforming to MIL-B-20390 with a nominal thickness of 0.024 inch.

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* 3.1.3.1.2.3 Fifth ply. The fifth ply shall be 0.001 inch minimum thickness, free aluminum foil, conforming to QQ-A-1876, overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029, or adhesive conforming to MMM-A-260.

3.1.3.1.2.4 Sixth ply. Material for sixth ply shall be waterproof duplex kraft paper 0.007 inch thick conforming to MIL-P-20311. The material shall be overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029.

3.1.3.1.2.5 Seventh ply. Material for seventh ply shall be asphalt-impregnated, kraft paper 0.006 inch thick conforming to MIL-P-20293. The material shall be overlapped not less than 1/4 inch and bonded with asphalt conforming to MIL-A-3029. A hairline of asphalt, not more than 1/32 inch in width and 1/64 inch in height, shall be allowable beyond the edge of the outer layer of asphalt impregnated kraft paper to which the asphalt is applied.

* 3.1.3.1.3 Ends. Single-body composite cans shall be fitted with one metal end at time of manufacture (see figures 1 and 7) and shall be capable of being sealed with a similar metal end after inserting contents. One loose metal end for final closure shall be furnished for each can.

* 3.1.3.2 Type II cans (telescopic type).

3.1.3.2.1 General. Type II cans shall be constructed of two multiple spirally wound tubes, one inside the other, fitted with metal ends and telescopic-type removable covers as shown in figure 2.

* 3.1.3.2.2 Can covers. The covers for type II (grade A and B) cans shall have a snug sliding fit on the neck tube. With the cover down, the cover shall move not more than one inch per second and shall require not more than 15 pounds for style b cans and not more than 10 pounds for style a cans to separate the cover from the can. Testing shall be as specified in 4.3.3.3.

3.1.3.2.3 Grade A cans. The required wall thickness of tubes for grade A cans shall be obtained by varying the number of plies of ammunition container board necessary to provide the thickness specified in Table III (see fig. 4).

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TABLE III

Dimensions, type II, grade A

Class (weight capacity)	Maximum diameter-inside of outer tube	Maximum Length (inside assembled)	Tube wall thickness	
			Cover and outer tube minimum	Neck tube minimum
	Inches	Inches	Inch	Inch
1	3	10	.065	.050
(Up to 5 lbs, inclusive)	5	20	.100	.100
	7-1/8	60	.100	.100
2	3	10	.065	.050
(Over 5 lbs. up to 10 lbs. inclusive)	5	20	.100	.100
	7-1/8	60	.100	.100
3	3	10	.065	.050
(Over 10 lbs. up to 15 lbs. inclusive)	5	20	.100	.100
	7-1/8	60	.100	.100

* 3.1.3.2.3.1 Cover and outer tubes. Cover and outer tubes shall be built up, from inside to outside, of one ply of greaseproof barrier material, conforming to MIL-L-45973, overlapped not less than 1/4 inch and the overlap glued, and the number of plies of ammunition container board necessary to obtain the specified wall thicknesses; and one outer ply of kraft wrapping paper with a minimum bursting strength of 48 points conforming to UU-P-268. The outer ply of kraft paper shall be wound with an overlapped joint not less than 1/4 inch wide and shall have the inner surface coated with adhesive conforming to MIL-A-13374.

* 3.1.3.2.3.2 Neck tubes. Neck tubes shall be built up, from inside to outside, of one inner ply of greaseproof barrier material, conforming to MIL-L-45973, and the number of plies of ammunition container board necessary to obtain the specified wall thickness. The inner ply of greaseproof paper shall be overlapped not less than 1/4 inch and the overlap glued with adhesive conforming to MIL-A-13374.

* 3.1.3.2.4 Grade B cans. The required wall thickness of tubes for grade B cans shall be obtained by varying the number of plies of ammunition container board necessary to provide the thickness specified in Table IV (see figure 5). Grade B can interiors shall be water-proof. Testing shall be as specified in 4.3.3.4.

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TABLE IV

Dimensions, type II, grade B

Class (Weight capacity)	Maximum diameter - inside of outer tube	Maximum length (inside assembled)	Tube wall thickness	
			Cover and outer tube minimum	Neck tube minimum
	Inches	Inches	Inch	Inch
1	3	10	.065	.050
(Up to 5 lbs. inclusive)	5	20	.100	.100
	7-1/8	60	.100	.100
2	3	10	.065	.050
(Over 5 lbs. up to 10 lbs. inclusive)	5	20	.100	.100
	7-1/8	60	.100	.100
3	3	10	.065	.050
(Over 10 lbs. up to 15 lbs. inclusive)	5	60	.100	.100
	7-1/8	20	.100	.100
4	3	10	.100	.100
(Over 15 lbs. up to 20 lbs. inclusive)	5	20	.100	.150
	7-1/8	60	.100	.150

* 3.1.3.2.4.1 Cover and outer tubes. Cover and outer tubes shall be built up, inside to outside, of one inner ply of greaseproof barrier material, conforming to MIL-A-45973, overlapped 1/4 inch and the overlap glued; the number of plies of ammunition container board necessary to obtain the specified wall thickness, one ply of .001 inch free aluminum foil, conforming to QQ-A-1876 overlapped not less than 1/4 inch and the overlap bonded with asphalt conforming to MIL-A-3029, or water-resistant adhesive in accordance with MMM-A-260; one ply of .007 inch waterproof kraft wrapping paper conforming to MIL-P-20311, and one outer ply of .006 to .007 inch asphalt-impregnated kraft paper conforming to MIL-P-20293. The waterproof duplex kraft paper shall be wound with overlapped joints not less than 1/4 inch wide and the inner surface shall be coated with asphalt conforming to MIL-A-3029. The outer ply of asphalt impregnated kraft paper shall be wound with an overlapped joint not less than 1/4 inch wide and shall have the inner surface coated with asphalt conforming to MIL-A-3029. A hairline of asphalt not more than 1/32 inch in width and 1/64 inch in height shall be allowable beyond the edge of the outer layer of asphalt impregnated kraft paper to which the asphalt is applied.

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* 3.1.3.2.4.2 Neck tubes. Neck tubes shall be built up, from inside to outside, of one inner ply of greaseproof barrier material conforming to MIL-L-45973, the number of plies of ammunition container board necessary to obtain the specified wall thickness, and one outer ply of .007-inch waterproof duplex kraft wrapping paper conforming to MIL-P-20311. The inner ply of greaseproof paper shall be wound with an overlapped joint not less than 1/4 inch wide and the overlap glued, and shall have the outer surface coated with adhesive conforming to MIL-A-13374. The outer ply of waterproof duplex kraft paper shall be wound with an overlapped joint not less than 1/4 inch wide and shall have the inner surface coated with asphalt conforming to MIL-A-3029. A hairline of asphalt, not more than 1/32 inch in width and 1/64 inch in height shall be allowable beyond the edge of the outer layer of asphalt impregnated kraft paper to which the asphalt is applied.

* 3.1.4 Design. Inside length and diameter of type I cans shall be as specified (see 6.2). Length and diameters of type II cans shall conform to Table III and IV as applicable, and shall be as specified by the procuring agency (see 6.2). A tolerance of ± 1/64 inch will be allowed on inside and outside diameters and ± 1/32 inch on specified length of both type I and type II cans.

3.2 Marking of cans. When required, marking of cans shall be as specified by the procuring agency (see 6.2).

3.3 Workmanship. The cans shall be free of surface scuffing, blisters, imperfect adhesion between layers, bleeding of asphalt at seams, checking ofpaper on inside and outside surfaces, ragged cuts, burns on the inside edges of the open ends of the cans, imperfect crimps, and other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Inspection records. Contractor's record of inspection shall be accurate, complete and available to the Government upon request. The contractor's records of inspection shall provide, as a minimum, the following information:

a. Contract order number.
b. Drawing and specification number.
c. Nomenclature.
d. Number of pieces inspected.
e. Individual characteristics inspected.
f. Conformance/nonconformance criteria.
g. Description and quantity of defects.
h. Results of inspection (e.g. identification of conforming, and of nonconforming product). Actual inspection and test results shall be recorded when actual measurements and tests were the basis for conformance/nonconformance decisions.
i. Date of inspection.
j. The following shall be included when sampling is used.

(1) Sampling plan.
(2) Acceptable Quality Levels (AQLs) used.
(3) Lot size and number.
(4) Sample size.

* 4.1.2 Certified test report (CTR). The contractor shall make available to the Government, a certified test report for each lot of composite cans by lot number prior to acceptance. Certified test reports are not required for commercial items. This test report is in addition to, and not in lieu of, any rights of the Government under this contract or law. A CTR may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of the contract item(s). As a minimum, the report shall contain the following:

a. Name of company and date.
b. Contract number or purchase order number, national stock number and drawing number.
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment will be given.
d. All inspections and tests required by contract (i.e., material, process, performance, functional, etc.) shall be recorded in test reports. These reports shall identify each lot, submitted for acceptance by lot number, the specification or drawing, revision and date, grade or type as applicable, number of specimens tested, specific characteristics and requirements, and actual results obtained.
e. Reports of the raw material producer's chemical and physical analysis.

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f. A statement, as follows, certifying that material meets all requirements of the contract:

"The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: All of the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not limited to the drawings and specifications); the inspection and test results, and the analysis appearing herein are true and accurate; and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for such payment."

g. Signature and title of certifying official.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Composite cans.

4.3.2.1.1 Sampling. Sampling shall be in accordance with MIL-STD-105, inspection level I.

4.3.2.1.2 Classification of defects. Classification of defects and acceptable quality levels (AQL's) shall be as follows:

Characteristic	Applicable Paragraph	AQL	Inspection Method
1. Cracked or split ends	3.1.2.1	1.5	Visual
2. Improper assembly	3.1.2, 3.1.3	1.5	Visual
3. Incorrect dimensions		1.5	SME*
4. Incorrect number of plies		1.5	Visual
5. Improper marking	3.2	4.0	Visual
6. Workmanship	3.3	4.0	Visual

*Standard Measuring Equipment

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4.3.2.2 Preparation for delivery. Sampling shall be in accordance with level S4 of MIL-STD-105. The acceptable quality level shall be 4.0 percent defective. One examination shall be made to determine that preservation, packing and marking comply with the requirements of Section 5 of this specification. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. A separate examination shall be performed for closure defects on shipping containers fully prepared for delivery. Sample unit shall be one fully prepared container. Inspection level and AQL shall be as stated above.

4.3.3 Testing.

* 4.3.3.1 Sampling. Unless otherwise specified, the sample size for testing shall be a quantity equal to 5 percent of the sample required in 4.3.2.1.1 but not less than 4 each, grade A cans or 6 each, grade B cans. Test results shall include all values upon which results are based and there shall be no failure to meet the sample unit or lot average requirements. When any specimen fails to pass any of the test, new samples shall be taken for re-testing. The sample for retesting shall contain twice as many specimens as for the first test. Failure of any of these samples shall be cause for rejection of the lot. Rejected lots may be resubmitted for sampling after all defective units are removed or defects corrected. Sampling of the resubmitted lot shall be twice as many specimens as for the first test. Failure of a sample shall cause final rejection of the lot.

4.3.3.2 Metal end stripping test. Not less than four cans or two-thirds of the sample taken in accordance with 4.3.3.1, whichever is larger, shall be tested for conformance to 3.1.2.2 using the test method specified in 4.4.1. Failure to meet the requirement shall be cause for rejection of the represented lot.

4.3.3.3 Can cover testing (type II only). Can covers shall be tested for fit (see 3.1.3.2.2) using the test method specified in 4.4.2. Failure to meet the requirement shall be cause for rejection of the represented lot.

* 4.3.3.4 Water proofness testing (Grade B cans only). Not less than 2 cans, or one-third of the sample specified in 4.3.3.1 shall be tested for water proofness (see 3.1.3.2.4) using the test method specified in 4.4.3. Failure to meet the requirement shall be cause for rejection of the represented lot.

* 4.3.4 Inspection equipment.

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* 4.3.4.1 Acquisition, calibration, maintenance and disposition. Unless otherwise specified (see 6.2), the responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.3.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetetive measurements to an accuracy of 10 percent of the total tolerance of characteristic being inspected.

* 4.4 Test methods.

* 4.4.1 Metal end stripping test. Can body assemblies shall be prepared by cutting the can tube (if necessary) to provide sections of proper length to be held securely in a testing fixture similar to that shown in figure 8. If other tubes, supports, or parts have been made a requirement by contract or drawing, such parts shall be removed from the can tube, for purpose of this test, without injury to the crimp of the metal end. Can bodies held in the fixture shall be tested in a testing machine or static testing arrangement satisfactory to the inspector. The load required to separate the metal end from the tube shall be determined by applying a load in the axial direction. When testing machine is used, rate of travel of head shall be 0.25 $\pm$ 0.05 inch per minute. All samples tested shall be considered unserviceable after test and shall not be returned to the lot.

4.4.2 Can cover test (type II only). The cover shall be rotated to the loosest closed position and the cans held in a vertical position with cover end down. Any movement and the rate of movement shall be noted to determine compliance with 3.1.3.2.2. If the cover fails to move when the can is in this position, a weight of 10 pounds for style a cans, or 15 pounds for style b cans, shall be suspended from the cover to determine compliance with 3.1.3.2.2.

* 4.4.3 Water proofness test (Grade B cans only). Properly sealed cans shall be conditioned at ambient conditions for a minimum of 4 hours prior to performing the water proofness test. The time that the cans and all fabricating materials have been maintained at ambient conditions prior to or during the fabrication period may be considered as part of the conditioning time. The cans shall then be tested by total immersion 1 inch below the surface of water at a minimum of 40 degrees Fahrenheit (F) for 15 minutes. After removal, the cans shall be opened and inspected for evidence of moisture.

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5. PACKAGING

5.1 Preservation (Type I metal ends only). (See 6.2).

* 5.1.1 Level A. Metal ends for Type I composite cans shall be preserved in one of the following containers: Fiberboard boxes conforming to PPP-B-636, class weather-resistant or composite cans conforming to MIL-C-3955. Closure shall be in accordance with the appendix to the applicable specification.

* 5.1.2 Level B. Level B preservation shall be the same as Level A. Commercial containers may be used in lieu of the specified container providing it is equivalent in protection and in weight and dimensional limitations. For Navy procurement, metal ends for type I composite cans shall be preserved in a manner that will afford adequate protection against deterioration or physical damage during shipment from the supply source to the first receiving activity.

* 5.1.3 Commercial. Commercial preservation shall conform to Fed. Std. No. 356.

5.2 Packing (see 6.2).

* 5.2.1 Level A. Composite cans shall be packed in shipping containers conforming to PPP-B-601 overseas type or PPP-B-621 class 2, in quantities as specified (see 6.2). The box closure and reinforcement shall be in accordance with the appendix of the applicable box specification. Metal ends for type 1 composite cans, unit packed as specified in 5.1.1 in a quantity equal to the number of type 1 composite cans, shall be packed in each exterior container. For Navy procurement PPP-B-636, class weather resistant and PPP-B-640 class 2 boxes are acceptable for a Level A pack. The box closure and reinforcement shall be in accordance with the appendix of the applicable box specification using tape and nonmetallic banding only.

5.2.2 Level B. Level B shipping containers shall conform to PPP-B-636, class weather-resistant or PPP-B-640, class 2.

* 5.2.3 Commercial. Commercial packing shall conform to Fed. Std. No. 356.

* 5.3 Marking. (Level A and B). Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Commercial. Marking shall be in accordance with Fed. Std. No. 356.

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6. NOTES

6.1 Intended use. Cans produced under this specification are intended for use as packaging containers for items requiring physical, waterproof, or moistureproof protection.

6.1.1 Cans conforming to grade B of this specification, when properly closed and sealed, meet the requirements of the submethods of MIL-P-116 as follows:

a. Type I - Submethods IA-13, IC-4, IIf.
b. Type II - Submethod IC-4.

6.1.2 In addition, cans conforming to this specification are suitable for use as intermediate containers for small unit packs.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, grade, style, and class (see 1.2).
c. Color of paint for ends (see 3.1.2.1).
d. Inside diameter and length (see 3.1.4).
e. Special marking (see 3.2).
f. Responsibility for acquisition, calibration, maintenance and disposition of inspection equipment (see 4.3.4.1).
g. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1 and 5.2).
h. Quantity to be packed in container (see 5.2.1).

* 6.3 Supersession data. Cans covered by MIL-C-5405, MIL-C-12147(ORD) and MIL-C-12804(ORD) have been previously superseded by the type of cans in this specification as indicated below:

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MIL-C-3955	MIL-C-5405	MIL-C-12147(ORD)	MIL-C-12804(ORD)
Type I			
(single body)	...	...	...
Grade B	...	...	...
Style a	...	Class 1	...
Class 1	...	...	...
Class 2	...	...	...
Style b	...	Class 2	...
Class 1	...	...	...
Class 2	...	...	...
Type II			
(telescopic body)	...	...	...
Grade A	Grade I	...	Grade I
Style a	Style B	...	Style A
Class 1	Class 1	...	Class 1
Class 2	Class 2	...	Class 2
Class 3	Class 3	...	Class 3
Style b	Style A	...	Style B
Class 1	Class 1	...	Class 1
Class 2	Class 2	...	Class 2
Class 3	Class 3	...	Class 3
Grade B	Grade II	...	Grade III
Style a	Style B	...	Style A
Class 1	Class 1	...	Class 1
Class 2	Class 2	...	Class 2
Class 3	Class 3	...	Class 3
Class 4	Class 4	...	Class 4
Style b	Style A	...	Style B
Class 1	Class 1	...	Class 1
Class 2	Class 2	...	Class 2
Class 3	Class 3	...	Class 3
Class 4	Class 4	...	Class 4

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* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - OS
Air Force - 69

Review activities:
Army - MI, SM, EA, WV
Navy - EC, SH
Air Force - 80, 82, 84
Defense Nuclear Agency - DS
Defense General Supply Center - GS

User activities:
Navy - MC, YD, SA
Air Force - 70

Preparing activity:
Army - WC

Project Number:
8110-0231

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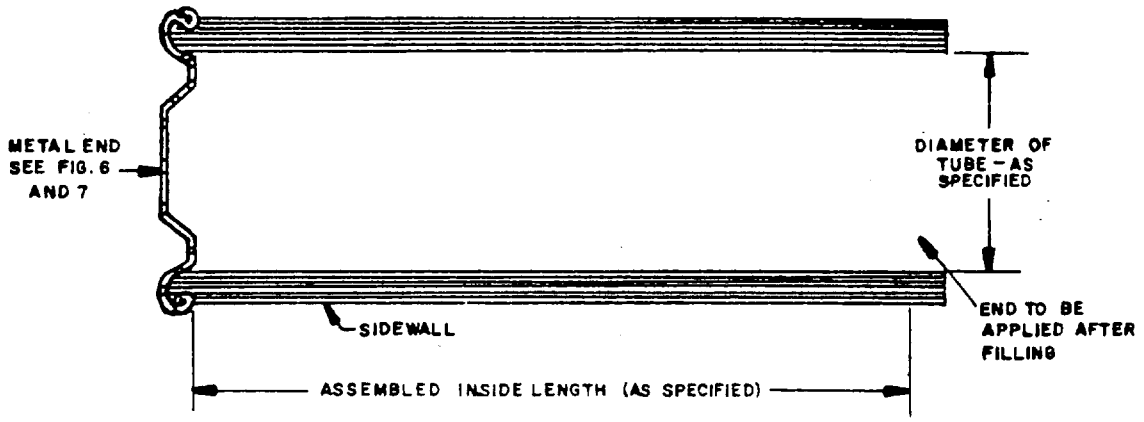
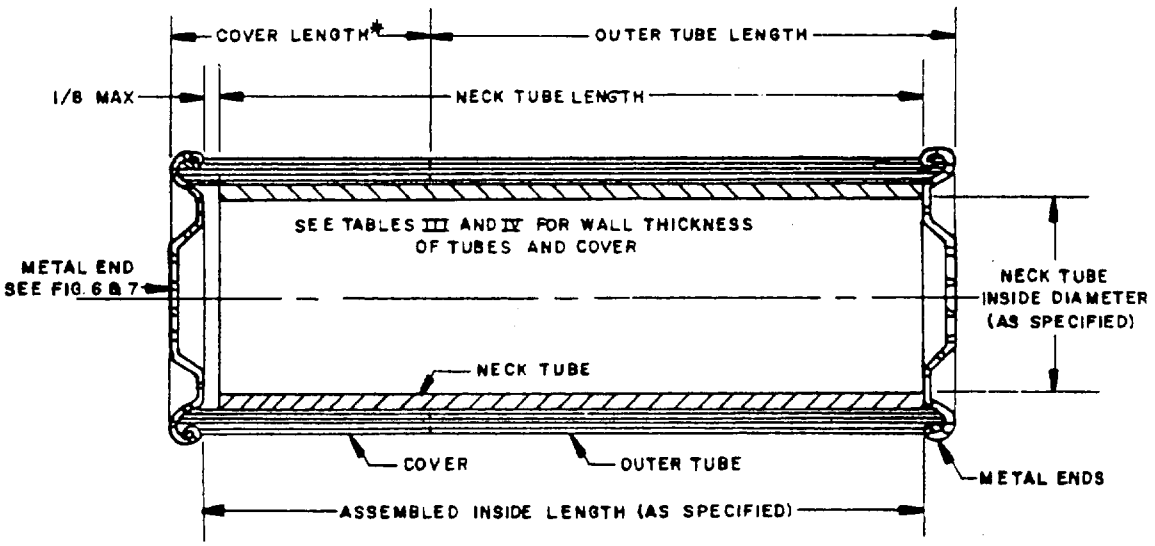


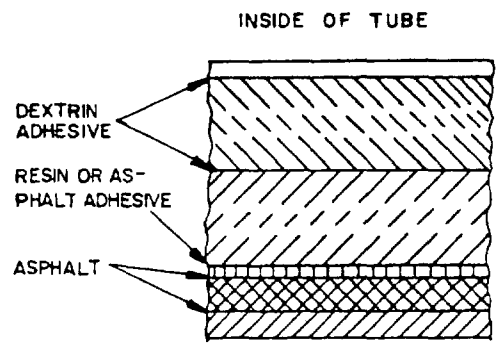
Figure 1.
Type I Can



* NOTE: Cover lengths are determined by the assembled inside length of the can. For cans 10 inches or over in length, the cover length shall be not more than 5 inches or less than 3 inches. For cans less than 10 inches in length, the cover length shall be not more than 1/2 or less than 1/3 of the can length.

Figure 2.
Type II Can

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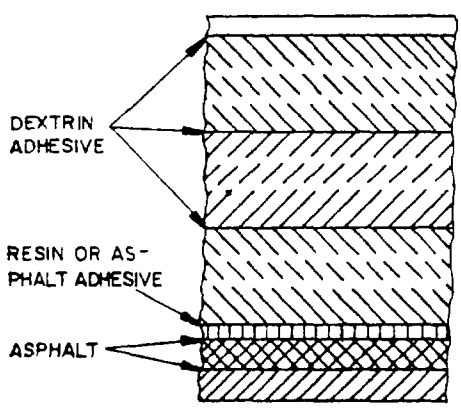
First Ply, Greaseproof Barrier Material

Second Ply, 0.020 inch Ammunition Container Board

Third Ply, 0.020 inch Ammunition Container Board

Fourth Ply, 0.001 inch Free Aluminum Foil
Fifth Ply, 0.007 inch Waterproof Duplex Kraft
Sixth Ply, 0.006 inch Asphalt-Impregnated Kraft

SECTION OF TUBE - TYPE I, GRADE B, STYLE a, CAN



First Ply, Greaseproof Barrier Material

Second Ply, 0.024 inch Ammunition Container Board

Third Ply, 0.024 inch Ammunition Container Board

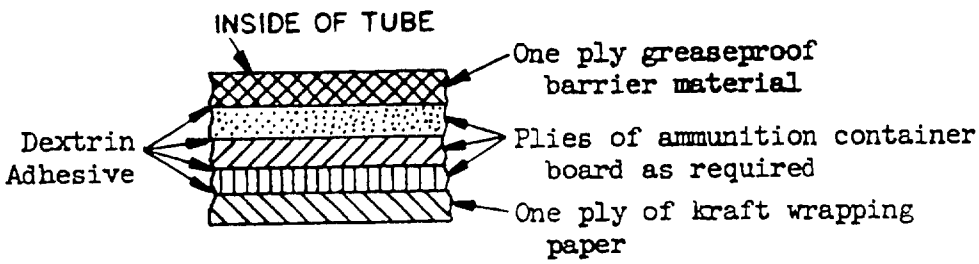
Fourth Ply, 0.024 inch Ammunition Container Board

Fifth Ply, 0.001 inch Free Aluminum Foil
Sixth Ply, 0.007 inch Waterproof Duplex Kraft
Seventh Ply, 0.006 inch Asphalt-Impregnated Kraft

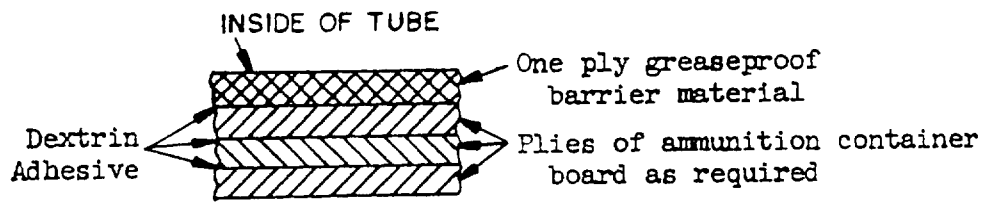
SECTION OF TUBE - TYPE I, GRADE B, STYLE b, CAN

Figure 3
Winding sequence for Tubes
Type I, Grade B, Styles a and b

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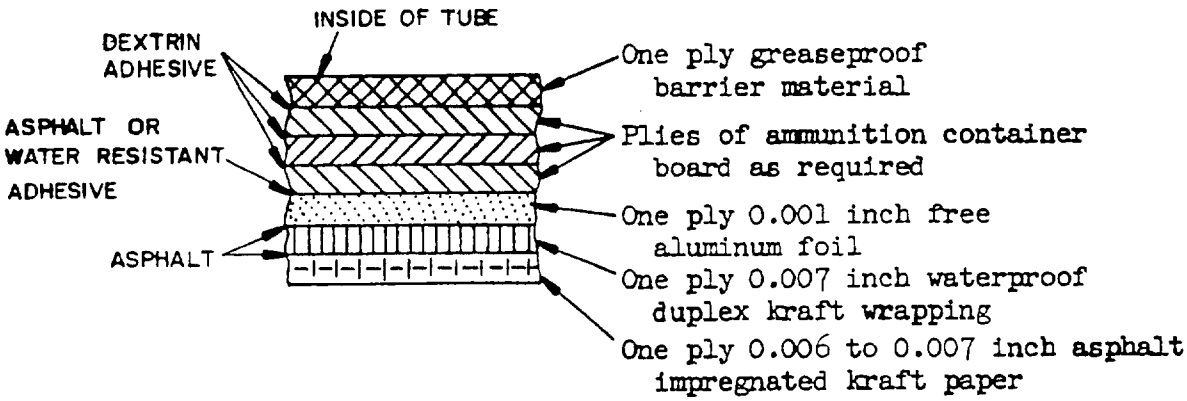
ENLARGED SECTION OF COVER & OUTER TUBE



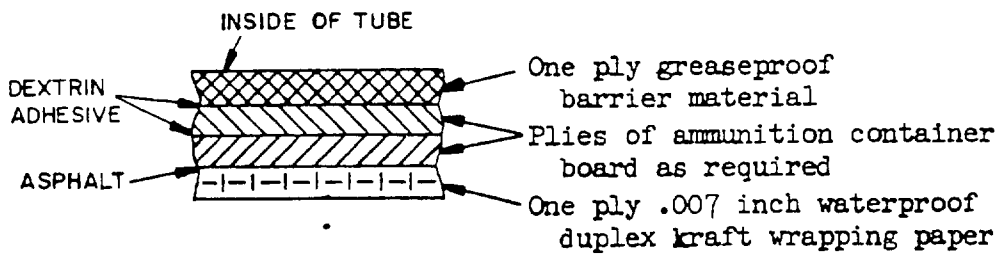
ENLARGED SECTION OF NECK TUBE

Figure 4. Construction of tube walls, Type II, grade A, styles a and b

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ENLARGED SECTION OF COVER & OUTER TUBE



ENLARGED SECTION OF NECK TUBE

Figure 5. Construction of tube walls
Type II, grade B, styles a and b

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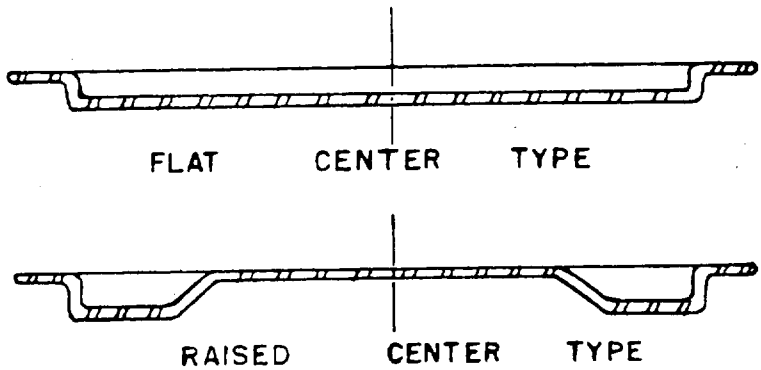
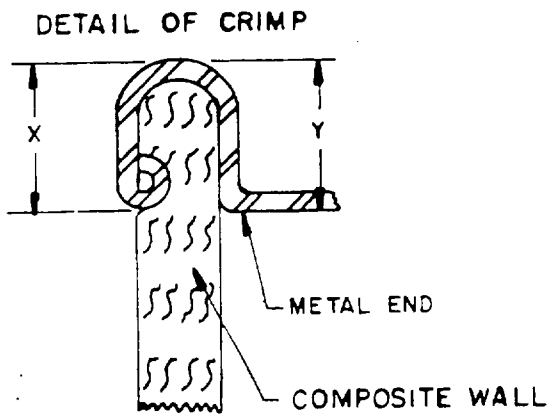


Figure 6. Ends for cans

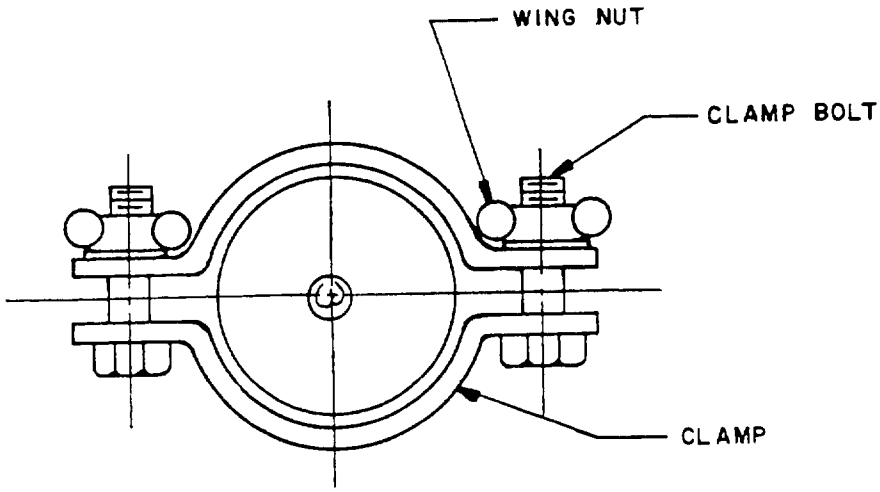


CRIMP DIMENSIONS

	MINIMUM INCHES	
	X	Y
Style a.		
Class 1.	3/32	1/8
Classes 2, 3, 4	5/32	3/16
Style b.		
Class 1.	5/32	3/16
Classes 2, 3, 4	15/64	9/32

Figure 7. Method of attaching ends

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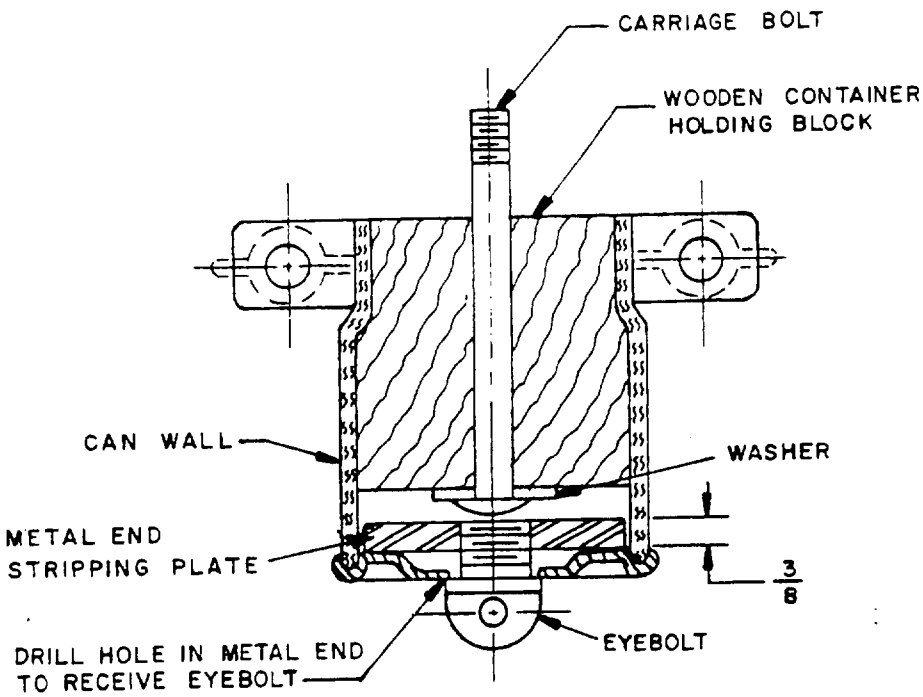


Figure 8
Fixture for stripping metal end from composite cans

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APPENDIX

10. SCOPE

10.1 This appendix covers the closure and sealing of spirally-wound composite cans.

20. APPLICABLE DOCUMENT

20.1 The following specification of the issue in effect on date of invitation for bids forms a part of this appendix.

SPECIFICATION

*	Federal MIL-T-43036	-	Tape; Pressure-Sensitive, Adhesive, Waterproof, For Packaging and Sealing

30. REQUIREMENTS

30.1 Type I composite cans. Type I composite cans shall be closed by means of a double crimp as specified in 3.1.2.2. Machines for closing these cans may usually be rented from manufacturer.

* 30.2 Type II composite cans. Type II composite cans shall be closed and sealed by the application of a strip of tape conforming to MIL-T-43036, type II, class 1, not less than 1-1/4 inches wide. A minimum length of tape 1-1/3 times the circumference of the can shall be centered over the joint of the body and the cover so that not less than 1/2 inch of tape adhers to the can or cover at any point.

40. QUALITY ASSURANCE PROVISIONS

40.1 Sampling. Sampling for examination of the closure and sealing of the cans shall be in accordance with MIL-STD-105. Lot acceptance or rejection shall be based on an AQL of 4.0 percent defective.

40.2 Examination. Determination shall be made that all cans presented for examination and testing of closure and sealing have been previously accepted under MIL-C-3955.

40.2.1 Type II cans. Type II cans shall be examined to determine that the type, grade, class, and style of cans are as specified in the commodity specification or procurement document, and that the joint of the cover and body is properly taped (see 30.2).

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40.2.2 Type I cans. Type I cans shall be examined for evidence of damage due to handling or during crimping operations and for conformance to type, grade, style, and class specified in the commodity specification or procurement document.

40.3 Tests (type I only). Two empty cans shall be closed by the same method used for closure of the lot and then subjected to the stripping test specified in 4.3.3.2 to determine the effectiveness of the crimping operation. If any sample does not pass this test, acceptance shall be suspended until it has been established that the defects have been corrected.

50. NOTES

50.1 Package engineering. When engineering a package which will employ this container, the designer should assure that the package or cushioning materials used (when required) will protect both the item and the container against rough handling.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-3955C; Cans, Composite, Spirally Wound

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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